

NAKSHATRA-X: Enterprise Mathematical & Algorithmic Blueprint

Complete Mathematical Formulations, Spectral Band Ratios, Linear Optimization & Telemetry Physics

1. AI Satellite Mineral Prospectivity Engine (Remote Sensing Mathematics)

The NAKSHATRA-X AI Prospectivity Engine processes Sentinel-2 multispectral satellite telemetry using key band ratio algorithms to identify subsurface manganese and iron ore alterations.

A. Sentinel-2 Band Spectral Indices

Index Name	Mathematical Formula	Target Geological Feature
Clay Minerals Index (CMI)	$CMI = B11 / B12 = SWIR1 / SWIR2$	Kaolinite, Alunite, & Hydrothermal Clay Alterations
Ferrous Minerals Index (FMI)	$FMI = (B12 / B8) + (B3 / B4)$	Ferrous Iron (Fe ²⁺) & Magnetite / Hematite Deposits
Iron Oxide Ratio Index (IORI)	$IORI = B4 / B2 = Red / Blue$	Gossans, Limonite, & Weathered Surface Ores
Manganese Alteration Index (MAI)	$MAI = (B11 * B4) / (B8 * B2)$	High-grade Mn Oxide / Gondite Alteration Zones

B. Logistic Regression Deposit Probability Formulation

The machine learning probability $P(Deposit = 1 | X)$ of a mineral deposit at coordinate (lat, lon) is modeled using the sigmoid transfer function:

$$\text{Logit Score } (z) = \beta_0 + \beta_1 \cdot CMI + \beta_2 \cdot FMI + \beta_3 \cdot IORI + \beta_4 \cdot MAI$$

$$\text{Deposit Probability } P(Y=1) = \sigma(z) = 1 / (1 + e^{(-z)})$$

C. Model Evaluation Metrics (Confusion Matrix Equations)

Metric	Formula	NAKSHATRA-X Value
Precision (P)	$P = TP / (TP + FP)$	92.4%
Recall (R)	$R = TP / (TP + FN)$	91.1%
F1-Score	$F1 = 2 * (P * R) / (P + R)$	91.7%
ROC-AUC Score	$AUC = \int_0^1 TPR(FPR) d(FPR)$	0.942 (94.2%)

2. Smart Ore Blending Optimization (Linear Programming Model)

To achieve target Manganese (%Mn) grade requirements while keeping unwanted impurities (%P, %SiO₂) below strict metallurgy limits, NAKSHATRA-X executes a Linear Programming (LP) optimization model:

Objective Function (Cost Minimization):

Minimize $Z = \sum (c_i * x_i)$ for $i = 1$ to N stockpiles

where c_i = Cost per tonne of stockpile i (₹/tonne), x_i = Tonnes extracted from stockpile i

Subject to System Constraints:

1. Total Target Tonnage: $\sum x_i = T_{\text{target}}$
2. Mn Grade Constraint (%Mn): $(\sum (g_{\text{Mn},i} * x_i)) / T_{\text{target}} \geq G_{\text{Mn},\text{min}}$
3. Phosphorus Cap (%P): $(\sum (g_{\text{P},i} * x_i)) / T_{\text{target}} \leq G_{\text{P},\text{max}}$
4. Silica Cap (%SiO₂): $(\sum (g_{\text{SiO}_2,i} * x_i)) / T_{\text{target}} \leq G_{\text{SiO}_2,\text{max}}$
5. Inventory Capacity Bound: $0 \leq x_i \leq \text{Available_Stockpile_Tonnes}_i$

3. Mine Digital Twin & Autonomous Telemetry Physics

For real-time vehicle positioning and underground hazardous gas telemetry, NAKSHATRA-X uses Haversine spherical geometry and gas safety threshold calculations.

A. Haversine Great-Circle Distance (GPS Coordinate Physics)

$$a = \sin^2(\Delta\phi / 2) + \cos(\phi_1) * \cos(\phi_2) * \sin^2(\Delta\lambda / 2)$$

$$c = 2 * \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$\text{Distance (d)} = R * c \text{ (Earth Radius } R = 6,371,000 \text{ meters)}$$

where ϕ_1, ϕ_2 are latitudes in radians, $\Delta\phi = (\text{lat}_2 - \text{lat}_1)$, $\Delta\lambda = (\text{lon}_2 - \text{lon}_1)$

B. Hazardous Underground Methane (CH₄) Gas Index Formula

$$\text{CH}_4 \text{ Concentration Percentage (\%)} = \text{Methane_PPM} / 10,000$$

$$\text{Hazard Index (\%)} = \min(100\%, (\text{CH}_4\% / 1.25\%) * 100)$$

Rule: If CH₄ ≥ 1.25% (Hazard Index = 100%), trigger automatic sirens & engine cutoff.

C. Heavy Haul Truck Fuel Consumption Equation

$$\text{Total Fuel (Liters)} = (\text{Idle_Rate} * t_{\text{idle}}) + (k_{\text{load}} * \text{Payload_Mass} * \text{Velocity} * t_{\text{hauling}})$$

where Idle_Rate = 4.5 L/hr, $k_{\text{load}} = 0.00018 \text{ L}/(\text{tonne} \cdot \text{km}/\text{h} \cdot \text{hr})$

4. Enterprise Security & ReDoS Complexity Boundaries

To protect the platform against cyberattacks, NAKSHATRA-X enforces algorithmic execution bounds:

1. ReDoS Backtracking Bound: String Length $L(s) \leq 256$ chars (Prevents $O(2^L)$ Regex Explosions)
2. Replay Nonce Window Expiry: $|t_{\text{current}} - t_{\text{nonce}}| \leq 300$ seconds (5 Minute Security Window)
3. Payload Size Limit: Content-Length $\leq 1,048,576$ bytes (1MB Max Payload Limit)